TED (15) 4024 Reg No……………..

(REVISION-2015) Signature………..

FOURTH SEMESTER DIPLOMA EXAMINATION IN MECHANICAL ENGINEERING

MODEL QUESTION

THERMAL ENGINEERING

Time: 3 hours Maximum Marks -100

(Use of steam tables& Mollier chart permitted)

I) Answer all questions in one or two sentences. Each question carries 2 marks.

1)State first law of thermodynamics.

2) Draw P-V diagram of diesel cycle.

3) Define Total fuel consumption.

4) Define thermal conductivity.

5) State any two uses of compressed air. (2X5=10)

PART-B

(Maximum Marks: 30)

II) Answer any five of the following questions.Each questions carries 6 marks

1) Explain thermodynamic system, boundary and surroundings.

2) A gas whose original pressure, volume and temperature were 140 KPa , 0.1 m^3 and 25^0C respectively is compressed to 700KPa and 60^0C, find the final volume of gas.

3) State the assumptions made in air standard cycles.

4) In an Otto cycle the temperature at beginning and the end of Isentropic compression are 316K and 596K respectively .Determine the compression ratio and air standard efficiency.

5) Explain Morse test.

6) Distinguish between wet steam, dry steam and super heated steam.

7) Define absorptivity, reflectivity and transmissivity. (5X6=30)

PART-C

Maximum Marks:60

(Answer one full question from each unit. Each question carries 15 marks)

UNIT- I

III) a) Derive the equations for work done and heat supplied for isothermal expansion process of gas. (7)

b) $0.12m^3$ of air at 1.5MPa and 1500^0C expands adiabatically to 175KPa. Find the final temperature and work done. (8)

OR

IV) a) Derive the equation for work done for adiabatic expansion of gas. (7)

b) $0.0001m^3$ of gas at $1000kN/m^2$ expands isothermally to a volume of $0.001m^3$, the initial temperature is 25^0C. Assume R=0.297kJ/kgK. Find 1) mass of gas,2) final pressure, 3) work done and

4) heat transferred. (8)

UNIT- II

V) a) Derive expression for air standard efficiency of Otto cycle with P-V diagram. (7)

b) Explain the working of four stroke petrol engine with schematic diagram. (8)

OR

VI) a) Calculate the air standard efficiency of a gas engine working on Otto cycle, bore and stroke of engine are 178mm and 305mm respectively. Clearance volume is 2200 $X10^3mm^{3.}$ (7)

b) Explain the working of two stroke diesel engine with suitable diagram. (8)

UNIT- III

VII) a) Derive expression for velocity of steam leaving a steam nozzle. (7)

b) Calculate total heat of 5kg of steam at absolute pressure of 8 bar and dryness fraction 0 .8.

Also calculate the heat to be supplied to convert this steam in to dry saturated steam. (8)

OR

VIII) a) Dry saturated steam enters the nozzle at 10 bar pressure and expands isentropically to 1 bar pressure. Determine the exit velocity of steam with mollier diagram. (7)

b) Explain the working of double acting steam engine with suitable figure. (8)

UNIT-IV

IX) a)A brick wall 250mm thick is faced with concrete 50mm thick. If the temperature of the exposed brick face is 30^0C and that of concrete is 5^0C.Determine the heat loss per hour through a wall of 10m long and 5m high. Also find out the interface temperature. (7)

b) Define mechanical efficiency , isentropic efficiency, isothermal efficiency and volumetric efficiency of an air compressor. (8)

OR

X) a) State and explain Fourier law of thermal conduction. (7)

b) Air is compressed through a pressure ratio of 10 from a pressure of 1 bar in one stage. Find the power required for a free air delivery of $3m^3$/minute. The compression follows the law $PV^{1.3}=C$ (8)